US IEC 60335-2-26

UGANDA STANDARD

First Edition
2006-11-14

Household and similar electrical appliances – Safety – Part 2-26: Particular requirements for clocks

Reference number

US IEC 60335-2-26:2002

© UNBS 2006

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Compliance with this standard does not, of itself confer immunity from legal obligations

A Uganda Standard does not purport to include all necessary provisions of a contract. Users are responsible for its correct application

© UNBS 2006

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilised in any form or by any means, electronic or mechanical, including photocopying and microfilm, without prior written permission from UNBS.

Requests for permission to reproduce this document should be addressed to

The Executive Director
Uganda National Bureau of Standards
P.O. Box 6329
Kampala
Uganda
Tel: +256 417 333 250/1/2
Fax: +256 414 286 123
E-mail: info@unbs.go.ug
Web: www.unbs.go.ug

© UNBS 2006 - All rights reserved
Licensed by UNBS to Uganda PVoC Programme-TEST
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

National foreword

Uganda National Bureau of Standards (UNBS) is a parastatal under the Ministry of Trade, Industry and Cooperatives established under Cap 327, of the Laws of Uganda, as amended. UNBS is mandated to co-ordinate the elaboration of standards and is

(a) a member of International Organisation for Standardisation (ISO),

(b) a contact point for the WHO/FAO Codex Alimentarius Commission on Food Standards, and

(c) the National Enquiry Point on TBT Agreement of the World Trade Organisation (WTO).

The work of preparing Uganda Standards is carried out through Technical Committees. A Technical Committee is established to deliberate on standards in a given field or area and consists of representatives of consumers, traders, academicians, manufacturers, government and other stakeholders.

Draft Uganda Standards adopted by the Technical Committee are widely circulated to stakeholders and the general public for comments. The committee reviews the comments before recommending the draft standards for approval and declaration as Uganda Standards by the National Standards Council.

This Uganda Standard US IEC 60335-2-26:2002 Household and similar electrical appliances – Safety – Part 2-26: Particular requirements for clocks, is identical with and has been reproduced from an International Standard IEC 60335-2-26:2002 Household and similar electrical appliances – Safety – Part 2-26: Particular requirements for clocks, and adopted as a Uganda Standard.

The committee responsible for this document is Technical Committee UNBS/TC 6, *Electro technology*.

Wherever the words, “International Standard" appear, they should be replaced by "Uganda Standard".

© UNBS 2006 - All rights reserved

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

INTERNATIONAL STANDARD

IEC 60335-2-26

Fourth edition
2002-10

Household and similar electrical appliances – Safety –

Part 2-26: Particular requirements for clocks

Appareils électrodomestiques et analogues – Sécurité –

Partie 2-26: Règles particulières pour les horloges

Reference number
IEC 60335-2-26:2002(E)

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Publication numbering

As from 1 January 1997 all IEC publications are issued with a designation in the 60000 series. For example, IEC 34-1 is now referred to as IEC 60034-1.

Consolidated editions

The IEC is now publishing consolidated versions of its publications. For example, edition numbers 1.0, 1.1 and 1.2 refer, respectively, to the base publication, the base publication incorporating amendment 1 and the base publication incorporating amendments 1 and 2.

Further information on IEC publications

The technical content of IEC publications is kept under constant review by the IEC, thus ensuring that the content reflects current technology. Information relating to this publication, including its validity, is available in the IEC Catalogue of publications (see below) in addition to new editions, amendments and corrigenda. Information on the subjects under consideration and work in progress undertaken by the technical committee which has prepared this publication, as well as the list of publications issued, is also available from the following:

- **IEC Web Site (www.iec.ch)**

- **Catalogue of IEC publications**

  The on-line catalogue on the IEC web site (http://www.iec.ch/searchpub/cur_fut.htm) enables you to search by a variety of criteria including text searches, technical committees and date of publication. On-line information is also available on recently issued publications, withdrawn and replaced publications, as well as corrigenda.

- **IEC Just Published**

  This summary of recently issued publications (http://www.iec.ch/online_news/justpub/jp_entry.htm) is also available by email. Please contact the Customer Service Centre (see below) for further information.

- **Customer Service Centre**

  If you have any questions regarding this publication or need further assistance, please contact the Customer Service Centre:

  Email: custserv@iec.ch
  Tel: +41 22 919 02 11
  Fax: +41 22 919 03 00

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

INTERNATIONAL STANDARD

IEC 60335-2-26

Fourth edition
2002-10

Household and similar electrical appliances – Safety –

Part 2-26: Particular requirements for clocks

Appareils électrodomestiques et analogues – Sécurité –

Partie 2-26: Règles particulières pour les horloges

© IEC 2002 — Copyright - all rights reserved

No part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from the publisher.

International Electrotechnical Commission, 3, rue de Varembé, PO Box 131, CH-1211 Geneva 20, Switzerland
Telephone: +41 22 919 02 11 Telefax: +41 22 919 03 00 E-mail: inmail@iec.ch Web: www.iec.ch

Commission Electrotechnique Internationale
International Electrotechnical Commission
Международная Электротехническая Комиссия

PRICE CODE L

For price, see current catalogue

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

CONTENTS

FOREWORD 3
INTRODUCTION 5

1 Scope 6
2 Normative references 6
3 Definitions 6
4 General requirement 7
5 General conditions for the tests 7
6 Classification 7
7 Marking and instructions 7
8 Protection against access to live parts 7
9 Starting of motor-operated appliances 7
10 Power input and current 7
11 Heating 7
12 Void 7
13 Leakage current and electric strength at operating temperature 7
14 Transient overvoltages 8
15 Moisture resistance 8
16 Leakage current and electric strength 8
17 Overload protection of transformers and associated circuits 8
18 Endurance 8
19 Abnormal operation 8
20 Stability and mechanical hazards 8
21 Mechanical strength 8
22 Construction 9
23 Internal wiring 9
24 Components 9
25 Supply connection and external flexible cords 9
26 Terminals for external conductors 9
27 Provision for earthing 9
28 Screws and connections 9
29 Clearances, creepage distances and solid insulation 10
30 Resistance to heat and fire 10
31 Resistance to rusting 10
32 Radiation, toxicity and similar hazards 10

Annexes 11

Bibliography 11

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

INTERNATIONAL ELECTROTECHNICAL COMMISSION

HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES – SAFETY –

Part 2-26: Particular requirements for clocks

FOREWORD

1) The IEC (International Electrotechnical Commission) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of the IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, the IEC publishes International Standards. Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. The IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
2) The formal decisions or agreements of the IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested National Committees.
3) The documents produced have the form of recommendations for international use and are published in the form of standards, technical specifications, technical reports or guides and they are accepted by the National Committees in that sense.
4) In order to promote international unification, IEC National Committees undertake to apply IEC International Standards transparently to the maximum extent possible in their national and regional standards. Any divergence between the IEC Standard and the corresponding national or regional standard shall be clearly indicated in the latter.
5) The IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with one of its standards.
6) Attention is drawn to the possibility that some of the elements of this International Standard may be the subject of patent rights. The IEC shall not be held responsible for identifying any or all such patent rights.

This part of International Standard IEC 60335 has been prepared by IEC technical committee 61: Safety of household and similar electrical appliances.

This fourth edition cancels and replaces the third edition published in 1994. It constitutes a technical revision.

The text of this part of 60335 is based on the following documents:

FDIS	Report on voting
61/2203/FDIS	61/2284/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This part 2 is to be used in conjunction with the latest edition of IEC 60335-1 and its amendments. It was established on the basis of the fourth edition (2001) of that standard.

NOTE 1 When “Part 1” is mentioned in this standard, it refers to IEC 60335-1.

This part 2 supplements or modifies the corresponding clauses in IEC 60335-1, so as to convert that publication into the IEC standard: Safety requirements for electric clocks.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

When a particular subclause of Part 1 is not mentioned in this part 2, that subclause applies as far as is reasonable. When this standard states “addition”, “modification” or “replacement”, the relevant text in Part 1 is to be adapted accordingly.

NOTE 2 The following numbering system is used:

– subclauses, tables and figures that are numbered starting from 101 are additional to those in Part 1;

– unless notes are in a new subclause or involve notes in Part 1, they are numbered starting from 101, including those in a replaced clause or subclause;

– additional annexes are lettered AA, BB, etc.

NOTE 3 The following print types are used:

– requirements: in roman type;

– *test specifications: in italic type;*

– notes: in small roman type.

Words in **bold** in the text are defined in Clause 3. When a definition concerns an adjective, the adjective and the associated noun are also in bold.

The committee has decided that the contents of this publication will remain unchanged until 2004. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

The following differences exist in the countries indicated below.

– 20.1: The requirement only applies to floor-standing clocks, which are tested at an angle of 8° (USA).

– 21.1: The impact tests are different (USA).

– 25.7: Lighter types of power supply cords are allowed (USA).

A bilingual version of this publication may be issued at a later date.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

INTRODUCTION

It has been assumed in the drafting of this International Standard that the execution of its provisions is entrusted to appropriately qualified and experienced persons.

This standard recognizes the internationally accepted level of protection against hazards such as electrical, mechanical, thermal, fire and radiation of appliances when operated as in normal use taking into account the manufacturer's instructions. It also covers abnormal situations that can be expected in practice.

This standard takes into account the requirements of IEC 60364 as far as possible so that there is compatibility with the wiring rules when the appliance is connected to the supply mains. However, national wiring rules may differ.

If an appliance within the scope of this standard also incorporates functions that are covered by another part 2 of IEC 60335, the relevant part 2 is applied to each function separately, as far as is reasonable. If applicable, the influence of one function on the other is taken into account.

This standard is a product family standard dealing with the safety of appliances and takes precedence over horizontal and generic standards covering the same subject.

An appliance that complies with the text of this standard will not necessarily be considered to comply with the safety principles of the standard if, when examined and tested, it is found to have other features that impair the level of safety covered by these requirements.

An appliance employing materials or having forms of construction differing from those detailed in the requirements of this standard may be examined and tested according to the intent of the requirements and, if found to be substantially equivalent, may be considered to comply with the standard.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES – SAFETY –

Part 2-26: Particular requirements for clocks

1 Scope

This clause of Part 1 is replaced by the following.

This International Standard deals with the safety of electric clocks having a **rated voltage** not more than 250 V.

NOTE 101 Examples of appliances that are within the scope of this standard are

– alarm clocks;

– spring-driven clocks with an electrically operated winding mechanism;

– clocks incorporating driving means other than motors.

As far as is practicable, this standard deals with the common hazards presented by appliances that are encountered by all persons in and around the home. However, in general, it does not take into account

– the use of appliances by young children or infirm persons without supervision;

– playing with the appliance by young children.

NOTE 102 Attention is drawn to the fact that

– for appliances intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary;

– in many countries additional requirements are specified by the national health authorities, the national authorities responsible for the protection of labour and similar authorities.

NOTE 103 This standard does not apply to

– battery-operated clocks;

– appliances intended exclusively for industrial purposes;

– appliances intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas);

– clocks having other functions, whether or not in combination with time indication, such as master control clocks and timers for cooking ranges, washing machines and similar appliances;

– clocks for "clocking-in" purposes;

– clocks incorporating electronic circuits only (IEC 60065).

2 Normative references

This clause of Part 1 is applicable.

3 Definitions

This clause of Part 1 is applicable except as follows.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

3.1.9 *Replacement:*

normal operation
operation of the clock under the following conditions

Spring-driven clocks having an electrically operated winding mechanism are operated as in normal use.

Other clocks are operated with the rotor locked.

4 General requirement

This clause of Part 1 is applicable.

5 General conditions for the tests

This clause of Part 1 is applicable.

6 Classification

This clause of Part 1 is applicable.

7 Marking and instructions

This clause of Part 1 is applicable.

8 Protection against access to live parts

This clause of Part 1 is applicable.

9 Starting of motor-operated appliances

This clause of Part 1 is not applicable.

10 Power input and current

This clause of Part 1 is applicable.

11 Heating

This clause of Part 1 is applicable.

12 Void

13 Leakage current and electric strength at operating temperature

This clause of Part 1 is applicable.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

14 Transient overvoltages

This clause of Part 1 is applicable.

15 Moisture resistance

This clause of Part 1 is applicable.

16 Leakage current and electric strength

This clause of Part 1 is applicable.

17 Overload protection of transformers and associated circuits

This clause of Part 1 is applicable.

18 Endurance

This clause of Part 1 is not applicable.

19 Abnormal operation

This clause of Part 1 is applicable except as follows.

19.7 *Addition:*

Spring-driven clocks, having an electrically operated winding mechanism incorporating capacitors or resistors to reduce the motor voltage, are operated with the rotor locked, capacitors or resistors being short-circuited one at a time.

20 Stability and mechanical hazards

This clause of Part 1 is applicable.

21 Mechanical strength

This clause of Part 1 is applicable except as follows.

Modification:

The impact energy is reduced to 0,20 J.

The blows are not applied to spindles for hands.

The blows are only applied to dial glasses if the clock fails to comply with the requirements of 8.1 with the dial glass removed.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

22 Construction

This clause of Part 1 is applicable except as follows.

22.35 *Addition:*

NOTE 101 Hands are not considered as being actuated in normal use, unless they have to be touched to alter the time setting.

23 Internal wiring

This clause of Part 1 is applicable.

24 Components

This clause of Part 1 is applicable.

25 Supply connection and external flexible cords

This clause of Part 1 is applicable except as follows.

25.3 *Modification:*

The connection to fixed wiring may be made before the clock has been fixed to its support unless the clock is classified at least IPX1.

25.5 *Addition:*

Type Z attachment is allowed.

25.7 *Addition:*

The **supply cords** may be flat non-sheathed cords (code designation 60227 IEC 42).

25.19 *Modification:*

Polyvinyl chloride insulated cord is allowed to be tied into a simple overhand knot around a smooth pin.

26 Terminals for external conductors

This clause of Part 1 is applicable.

27 Provision for earthing

This clause of Part 1 is applicable.

28 Screws and connections

This clause of Part 1 is applicable.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

29 Clearances, creepage distances and solid insulation

This clause of Part 1 is applicable.

30 Resistance to heat and fire

This clause of Part 1 is applicable except as follows.

30.2.2 Not applicable

31 Resistance to rusting

This clause of Part 1 is applicable.

32 Radiation, toxicity and similar hazards

This clause of Part 1 is applicable.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Annexes

The annexes of Part 1 are applicable.

Bibliography

The bibliography of Part 1 is applicable.

_____

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Standards Survey

The IEC would like to offer you the best quality standards possible. To make sure that we continue to meet your needs, your feedback is essential. Would you please take a minute to answer the questions overleaf and fax them to us at +41 22 919 03 00 or mail them to the address below. Thank you!

Customer Service Centre (CSC)

International Electrotechnical Commission
3, rue de Varembé
1211 Genève 20
Switzerland

or

Fax to: **IEC**/CSC at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

A Prioritaire

Nicht frankieren
Ne pas affranchir

Non affrancare
No stamp required

RÉPONSE PAYÉE

SUISSE

Customer Service Centre (CSC)
International Electrotechnical Commission
3, rue de Varembé
1211 GENEVA 20
Switzerland

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Q1 Please report on **ONE STANDARD** and **ONE STANDARD ONLY**. Enter the exact number of the standard: *(e.g. 60601-1-1)*

.....

Q2 Please tell us in what capacity(ies) you bought the standard *(tick all that apply).* I am the/a:

purchasing agent ❑
librarian ❑
researcher ❑
design engineer ❑
safety engineer ❑
testing engineer ❑
marketing specialist ❑
other.....

Q3 I work for/in/as a: *(tick all that apply)*

manufacturing ❑
consultant ❑
government ❑
test/certification facility ❑
public utility ❑
education ❑
military ❑
other.....

Q4 This standard will be used for: *(tick all that apply)*

general reference ❑
product research ❑
product design/development ❑
specifications ❑
tenders ❑
quality assessment ❑
certification ❑
technical documentation ❑
thesis ❑
manufacturing ❑
other.....

Q5 This standard meets my needs: *(tick one)*

not at all ❑
nearly ❑
fairly well ❑
exactly ❑

Q6 If you ticked NOT AT ALL in Question 5 the reason is: *(tick all that apply)*

standard is out of date ❑
standard is incomplete ❑
standard is too academic ❑
standard is too superficial ❑
title is misleading ❑
I made the wrong choice ❑
other .....

Q7 Please assess the standard in the following categories, using the numbers:
(1) unacceptable,
(2) below average,
(3) average,
(4) above average,
(5) exceptional,
(6) not applicable

timeliness.....
quality of writing.....
technical contents.....
logic of arrangement of contents .....
tables, charts, graphs, figures.....
other .....

Q8 I read/use the: *(tick one)*

French text only ❑
English text only ❑
both English and French texts ❑

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
.....

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

Licensed by UNBS to Uganda PVoC Programme-TUV
Order # NUMBER/Downloaded: 2024-04-19
Single-user licence only, copying and networking prohibited.

ISBN 2-8318-6662-6



ICS 13.120; 97.040.50

Typeset and printed by the IEC Central Office
GENEVA, SWITZERLAND

Licensed by ONBS to Qualitätsmanagement TÜV
Order # NUMBER/Downloaded, 2011-12-12
Single-user licence only, copying and networking prohibited.